

Area, packing density, whitespace/deadspace

With a legal packing in hand, score the whitespace

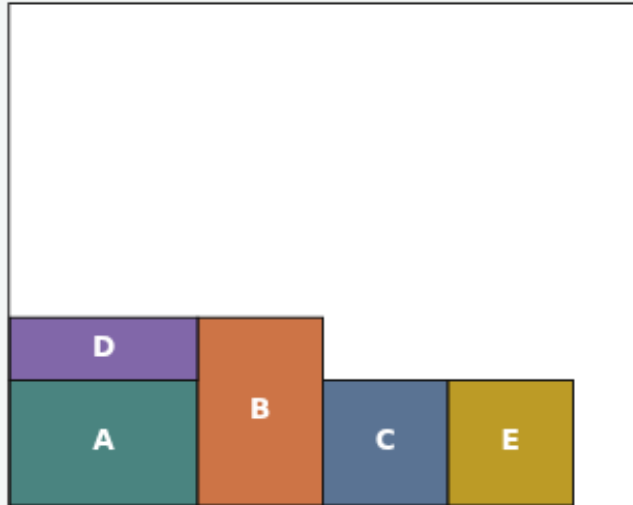
Module areas sum to 23

AREA & DEADSPACE

Module areas sum to 23

Step 1 / 5 · areas · module01-03-area-deadspace

10×8



Explanation

Add the five module areas: A is six, B is six, C is four, D is three, E is four. That is twenty-three units of silicon that must fit in the outline.

- $A 3 \times 2 = 6$
- $B 2 \times 3 = 6$
- $C 2 \times 2 = 4$
- $D 3 \times 1 = 3$
- $E 2 \times 2 = 4$

moduleAreaSum: 23

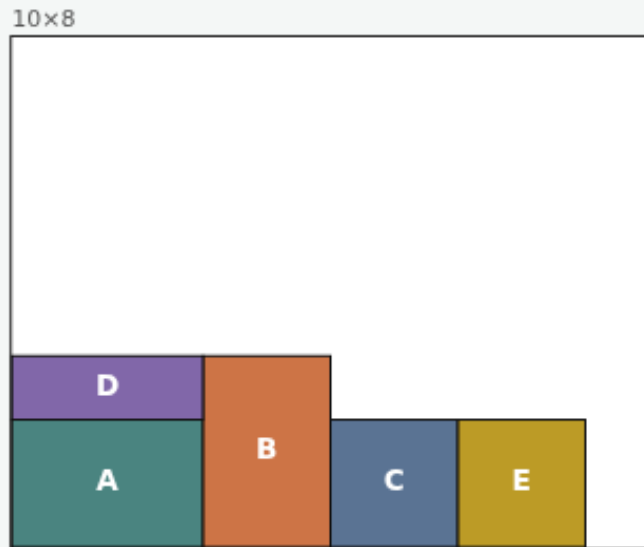
Module areas sum to 23

Outline area is 80

AREA & DEADSPACE

Outline area is 80

Step 2 / 5 · outline-area · module01-03-area-deadspace



Explanation

Ten times eight is eighty. That is the budget. Module area twenty-three leaves whitespace—deadspace—equal to fifty-seven.

- $\text{outlineArea} = W \times H = 80$
- $\text{deadspace} = 80 - 23 = 57$

```
outlineArea: 80  
deadspace: 57
```

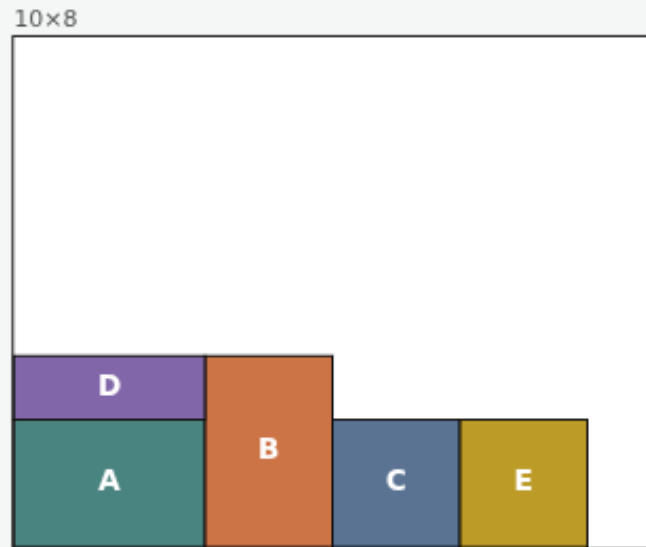
Outline area is 80

Density is 0.2875

AREA & DEADSPACE

Density is 0.2875

Step 3 / 5 · density · module01-03-area-deadspace



Explanation

Density is module area over outline area: twenty-three over eighty equals zero point two eight seven five. Whitespace fraction is zero point seven one two five.

- $\text{density} = 23/80 = 0.2875$
- $\text{whitespace} = 57/80 = 0.7125$
- Placement-independent for fixed sizes

density: 0.2875
whitespace: 0.7125

Density is 0.2875

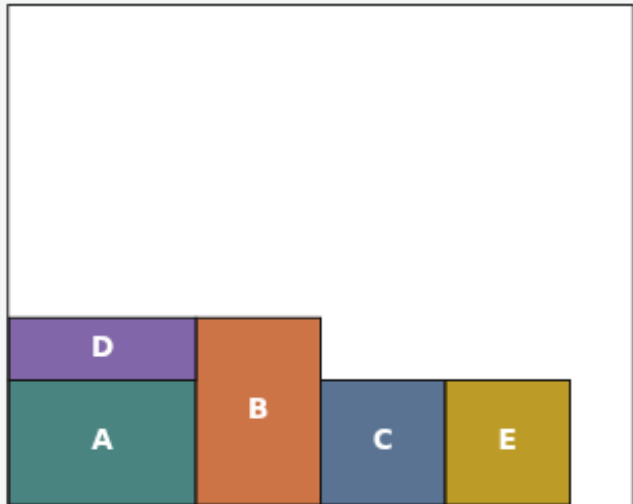
Metrics need a legal pack

AREA & DEADSPACE

Metrics need a legal pack

Step 4 / 5 · legal-metrics · module01-03-area-deadspace

10×8



Explanation

On the golden legal packing these numbers are honest. On an overflow packing the area math still computes, but the floorplan is invalid—do not celebrate density on illegal layouts.

- Golden packing is legal
- Deadspace 57 is the teaching golden
- Illegal packs still report area math

```
legal: true  
deadspace: 57  
density: 0.2875
```

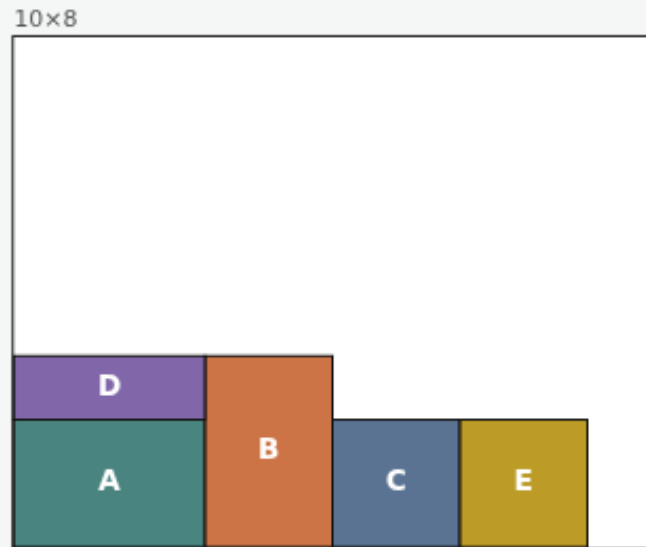
Metrics need a legal pack

Report area trio every time

AREA & DEADSPACE

Report area trio every time

Step 5 / 5 · takeaway · module01-03-area-deadspace



Explanation

Habit: print module area, outline area, deadspace, and density with every packing. Representations later change geometry; this trio stays the scoreboard.

- Always report 23 / 80 / 57 / 0.2875
- Density < 1 means whitespace remains
- Next: slicing polish that fits BB 9×3

golden: 23, 80, 57, 0.2875

Report area trio every time

Browser lab track

- Open area-deadspace, load the golden packing, and read the metrics panel

Implement track

- Implement `moduleAreaSum`, `outlineArea`, `deadspace`, and `density`
- Assert `deadspace` equals eighty minus twenty-three on the starter modules

Pitfalls

- Reporting density on an illegal pack

Your turn

- Print the area trio on every run
- Next lab builds a slicing polish whose bounding box is nine by three inside this outline